

EXP 10: Write a Python Program to implement the Expectation & Maximization (EM) Algorithm.

AIM

To implement the Expectation-Maximization (EM) Algorithm using Python for clustering.

SOFTWARE USED

1. Google Colab
2. Python 3.x
3. NumPy
4. Matplotlib
5. Scikit-learn

PROCEDURE

1. Import the required libraries.
2. Generate a sample dataset.
3. Create a Gaussian Mixture Model (GMM).
4. Train the model using the EM algorithm.
5. Predict cluster labels.
6. Plot the clustered data.

GOOGLE COLAB CODE

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.mixture import GaussianMixture

from sklearn.datasets import make_blobs

# Generate dataset

X, y = make_blobs(n_samples=300,

                  centers=3,

                  cluster_std=0.60,

                  random_state=0)

# Gaussian Mixture Model (EM)

gmm = GaussianMixture(n_components=3, random_state=0)

gmm.fit(X)

labels = gmm.predict(X)

# Plot clusters

plt.scatter(X[:,0], X[:,1], c=labels, cmap='viridis')

plt.title("Expectation Maximization Clustering")

plt.xlabel("Feature 1")

plt.ylabel("Feature 2")

plt.show()
```

OUTPUT



- A scatter plot is displayed showing three clusters identified using the Expectation-Maximization (EM) Algorithm.)

RESULT

Thus, the Expectation-Maximization (EM) Algorithm was successfully implemented using a Gaussian Mixture Model (GMM), and the data points were clustered into distinct groups.